

22050 Signals and linear systems in continuous time

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L01

Classification of signals and systems

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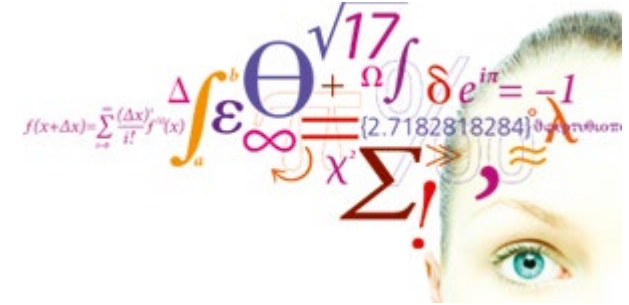
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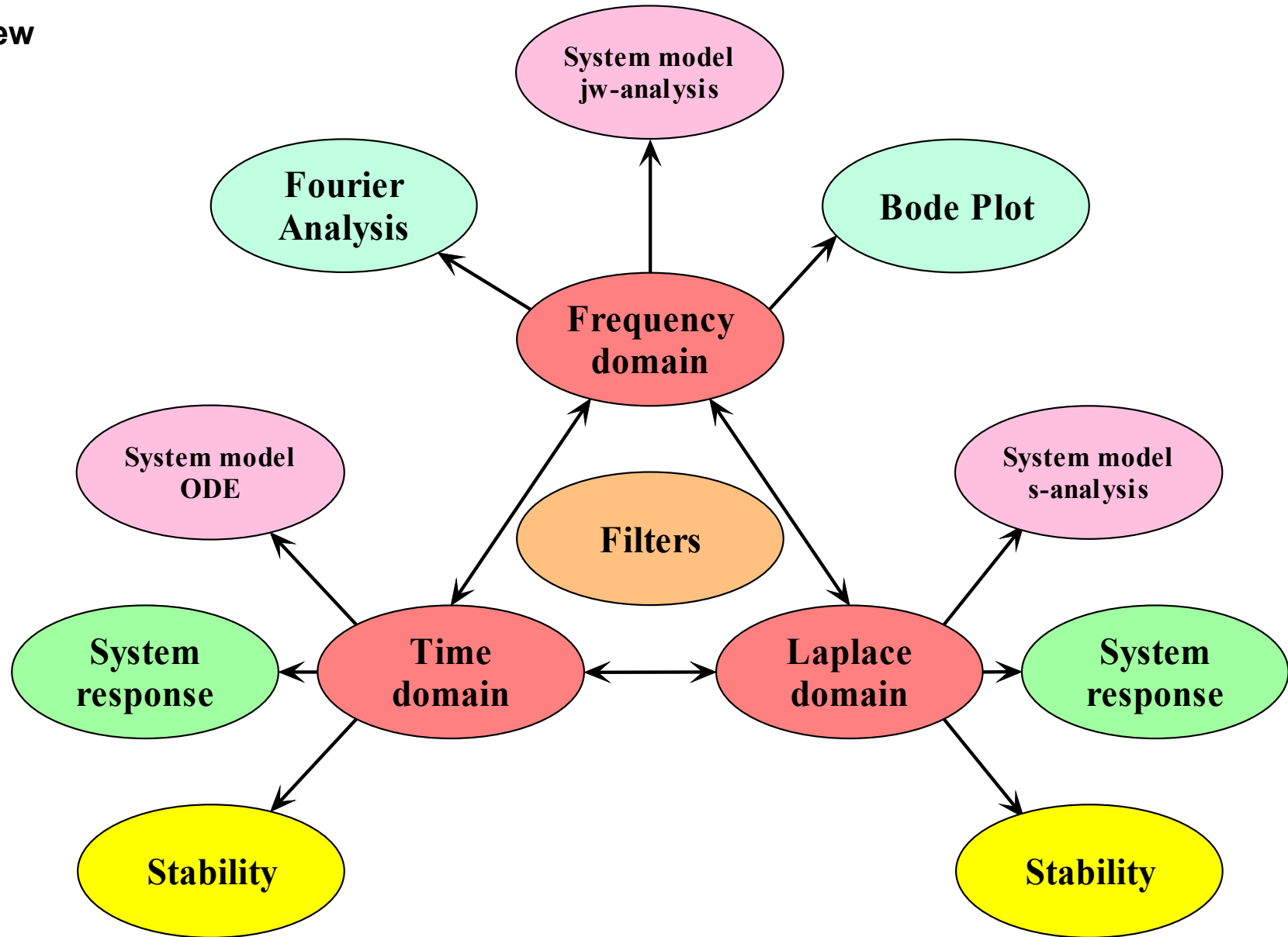
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- Lecture
 - Overview of "Signals and Systems part"
 - Classification of signals
 - Operations on signals
 - Special signals
 - Classification of systems
 - Overview of filter problems
- Exercises

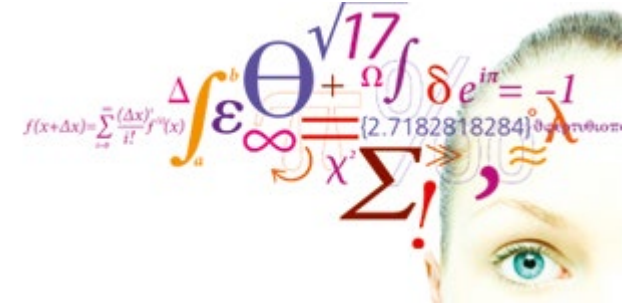


- System
 - Model
 - Response
 - Performance

- Domains
 - Time
 - Frequency
 - s-domain



- Lecture
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 - **Classification of signals**
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Video 2

Deterministic signal:

$$x(t) = \begin{cases} 0 & t < 0 \\ t^4 e^{-t^2/3} & t \geq 0 \end{cases}$$

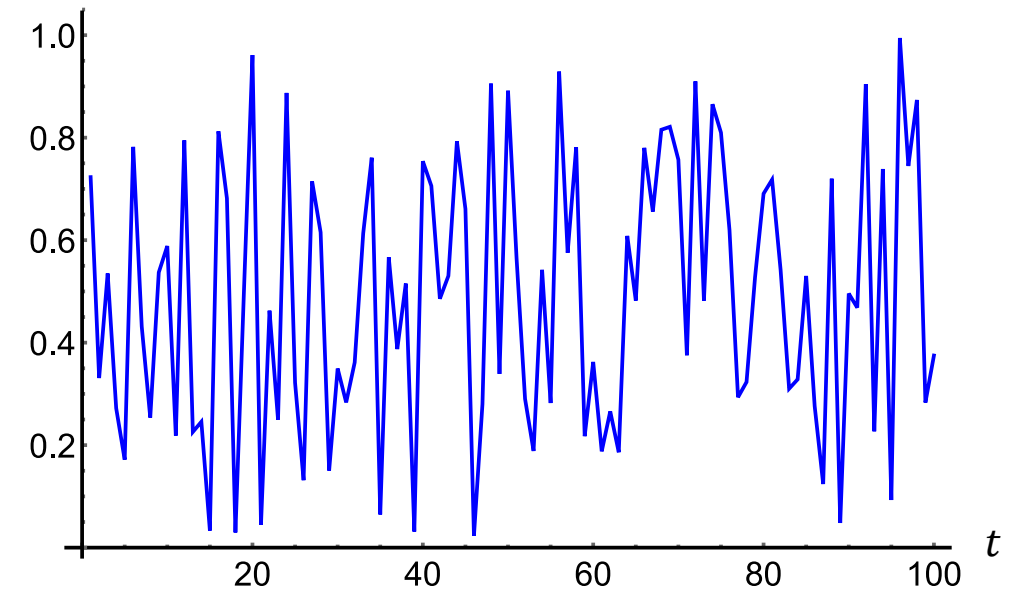
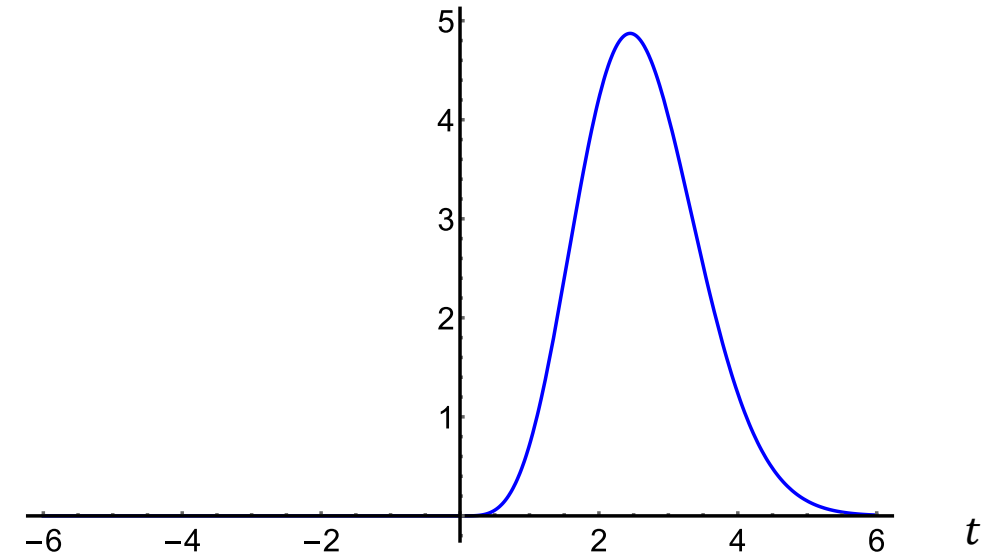
Can be expressed in mathematical equations.

Stochastic signals:

The process generating the signal is random.

No equations can be used to express the amplitude at any given time.

A deterministic signal has a known functional dependency of the independent variable, which often is time, but also could be space or some other quantity. Thus, for a deterministic time signal we can always tell at time t_0 what the signal amplitude will be at time t_1 even when $t_1 > t_0$.

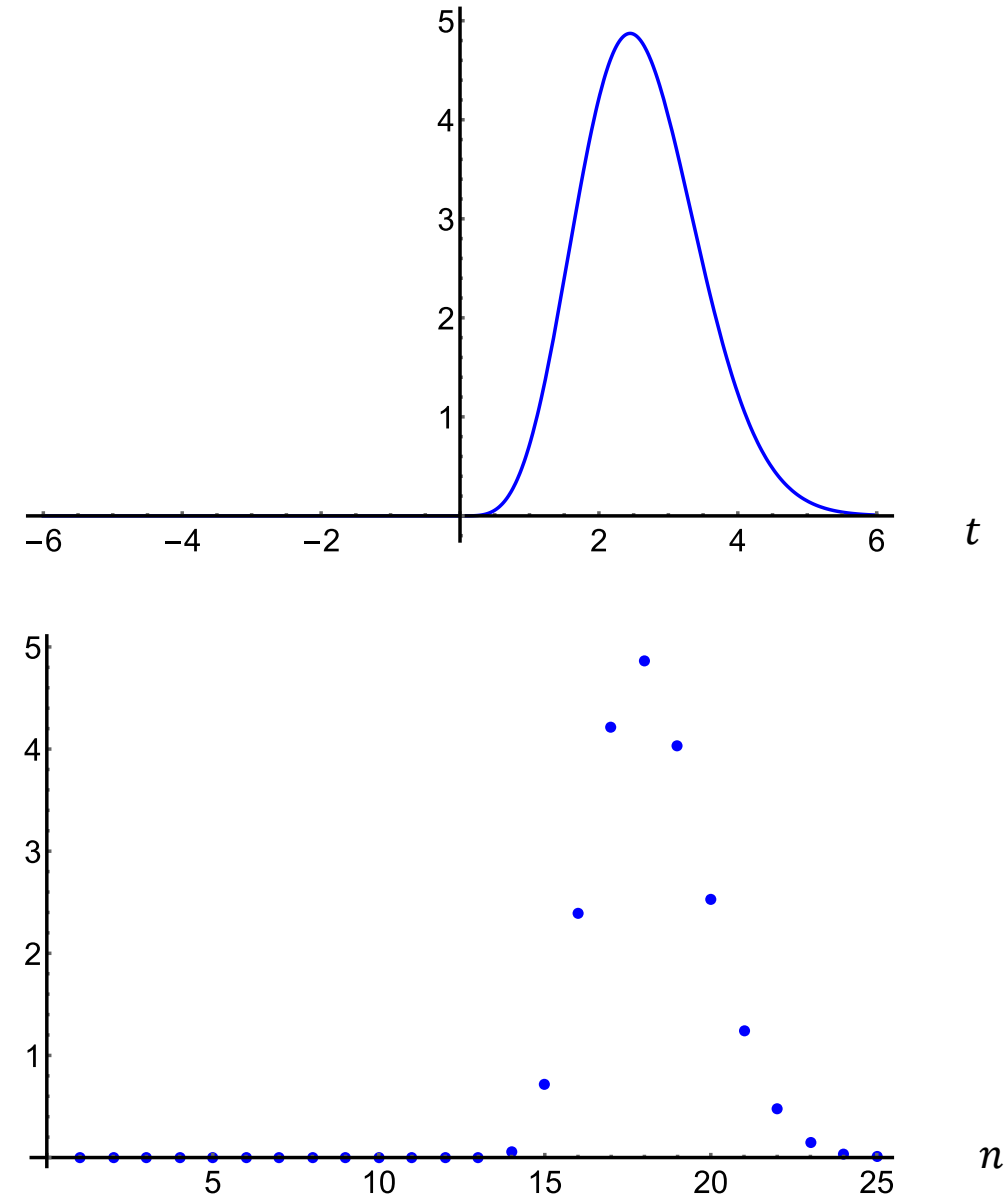


Continuous time versus discrete time signals

Continuous time signal: $x(t) = \begin{cases} 0 & t < 0 \\ t^4 e^{-t^2/3} & t \geq 0 \end{cases}$

Discrete time signals: $x(n\Delta t) = \begin{cases} 0 & n < 0 \\ (n\Delta t)^4 e^{-(n\Delta t)^2/3} & n \geq 0 \end{cases}$

Analog signals are continuous both in amplitude and time. Analog signals are also often called continuous-time (CT) signals. Discrete-time (DT) signals are defined only at certain (often constant) time intervals. The samples of DT signals are assumed to have continuous amplitude values. A **digital signal** is discrete both in time and in amplitude, i.e., the amplitude of each sample is represented by a finite number of bits.

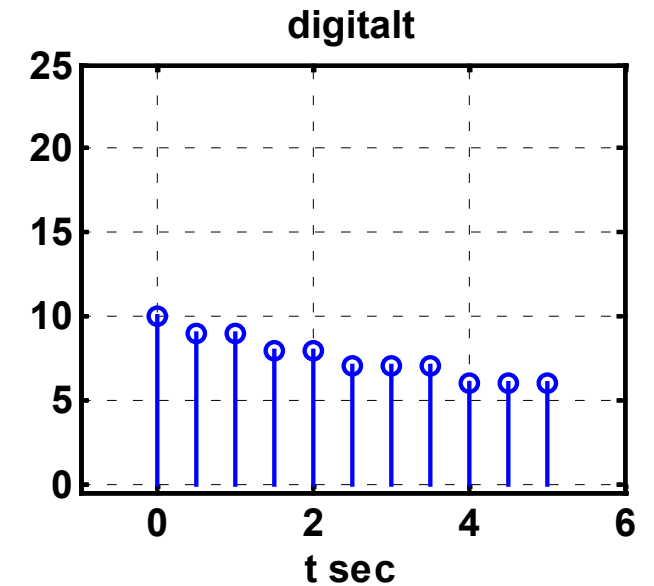
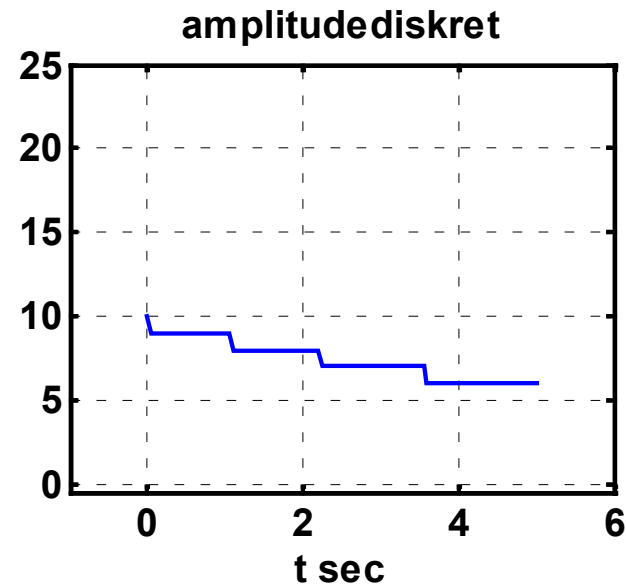
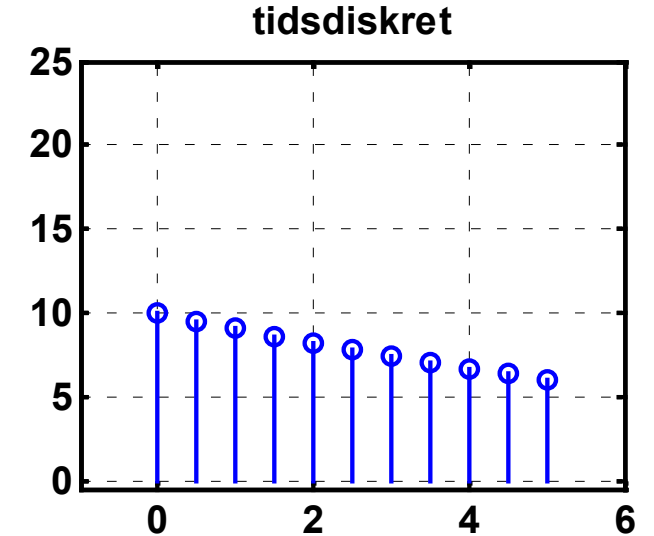
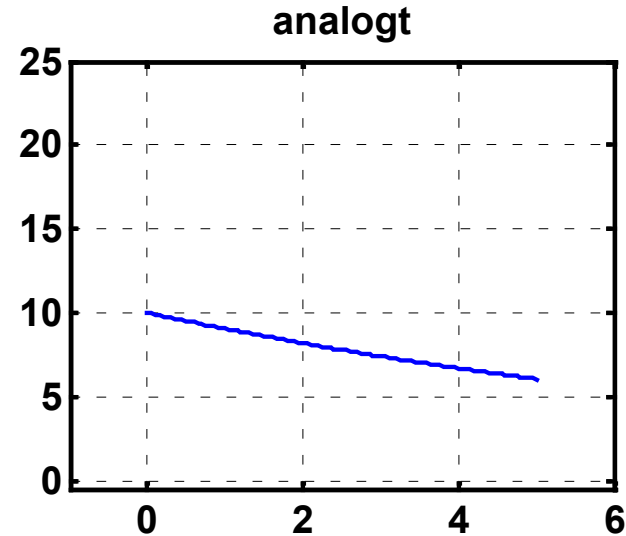


Analogue signals are continuous in both time and amplitude.

Digital signals are discrete in both time and amplitude.

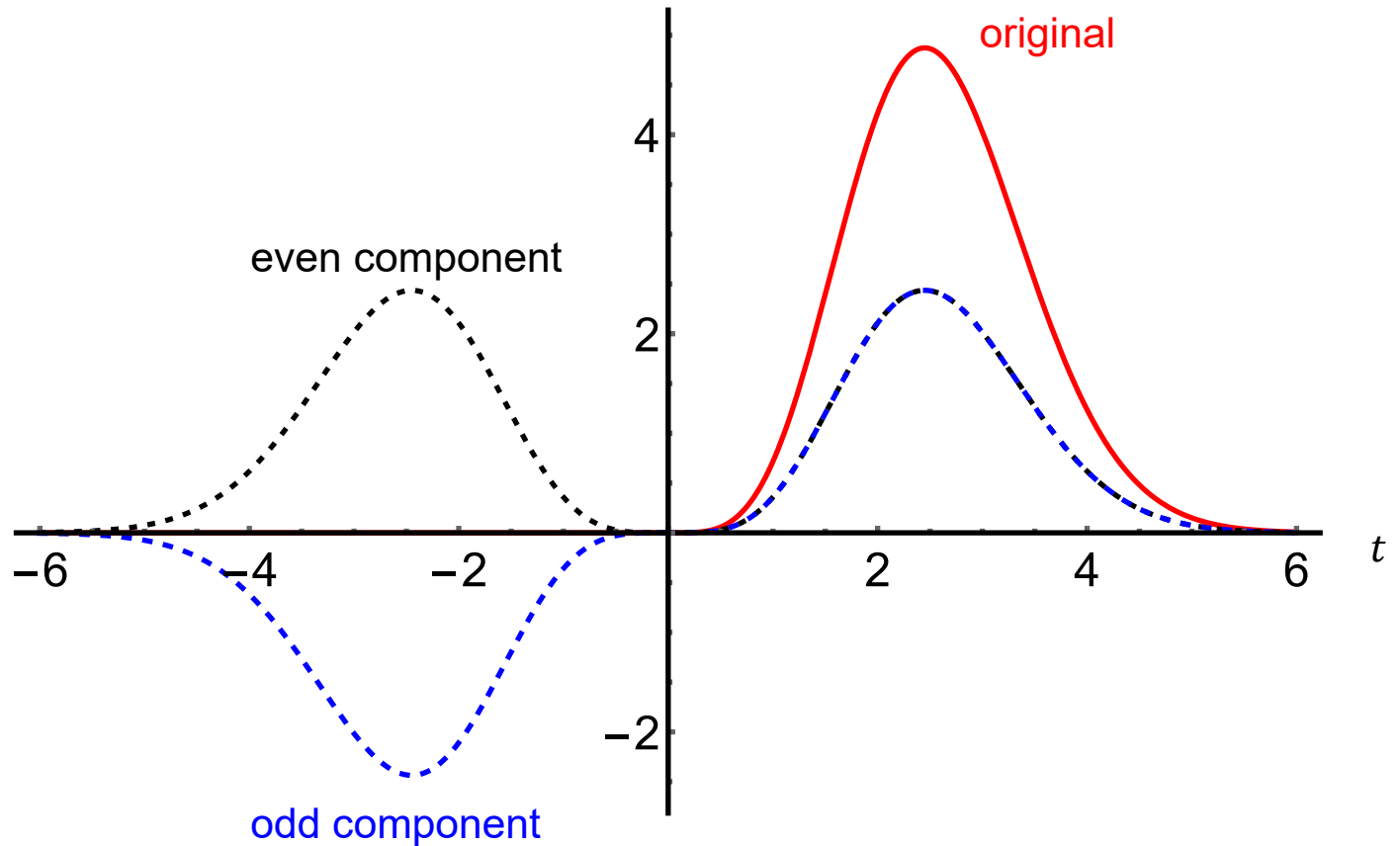
Digital samples are stored in binary form with a finite number of bits.

$$\Delta V_{LSB} = \frac{ADC \text{ Voltage range}}{2^N}$$



Even signal: $x_e(t) = \frac{x(t) + x(-t)}{2}$

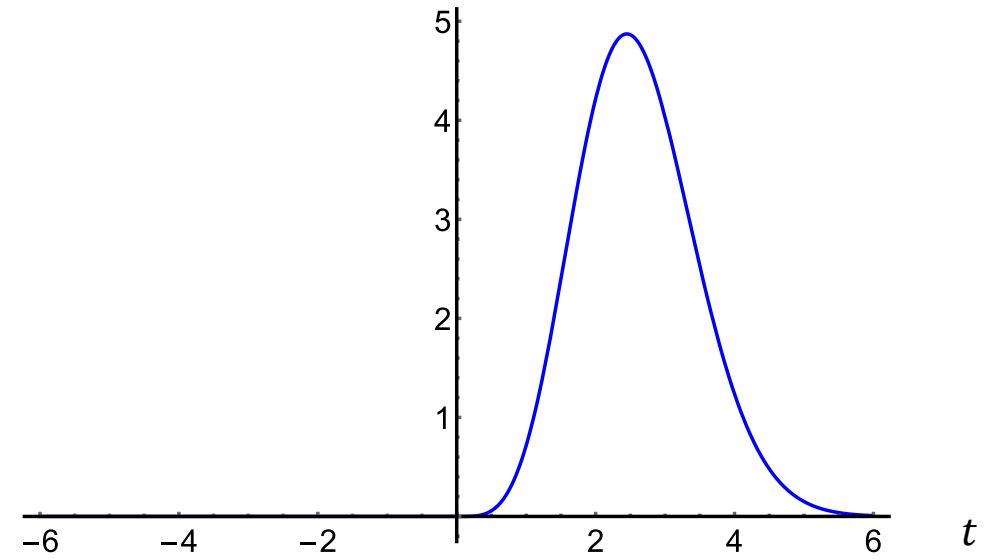
Odd signal: $x_o(t) = \frac{x(t) - x(-t)}{2}$



- A signal that is neither even nor odd still has an even component and an odd component.
- A signal that is even has zero odd component.
- A signal that is odd has zero even component.

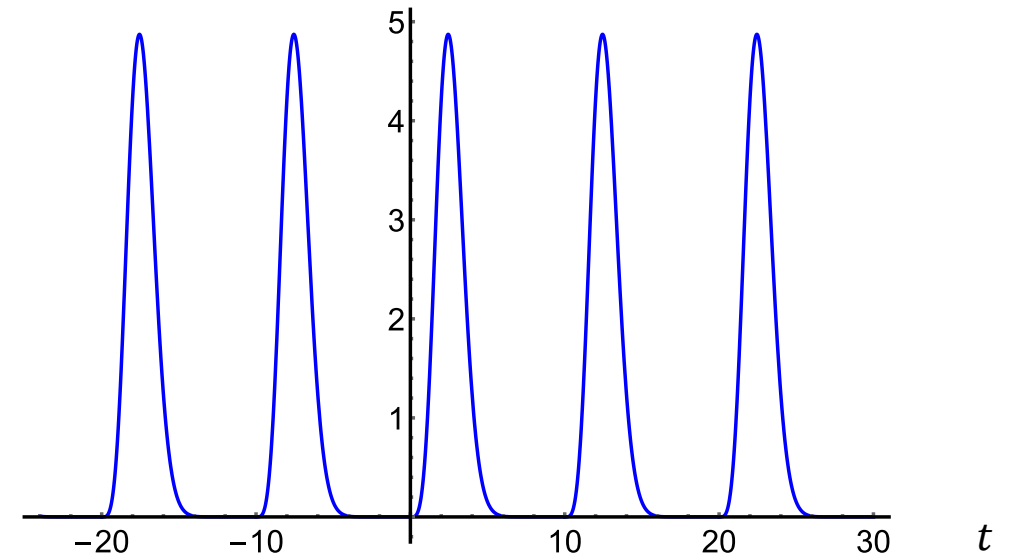
Aperiodic signal:

$$x(t) = \begin{cases} 0 & t < 0 \\ t^4 e^{-t^2/3} & t \geq 0 \end{cases}$$

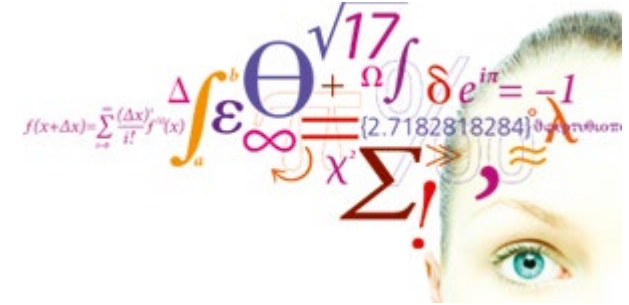


Periodic signal:

$$y(t) = \sum_{i=-3}^3 x(t - i \cdot T_0) \quad T_0 = 10$$



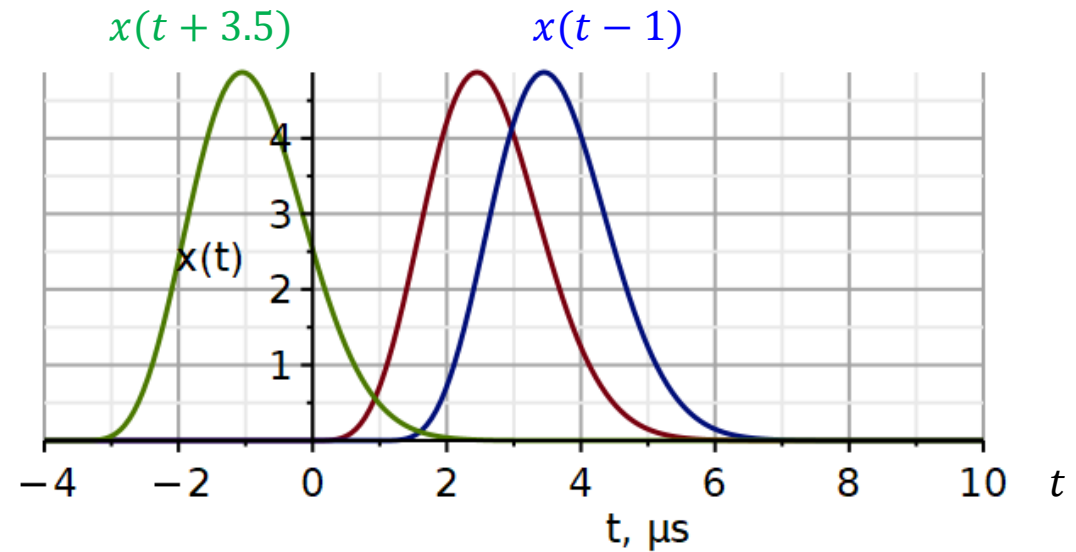
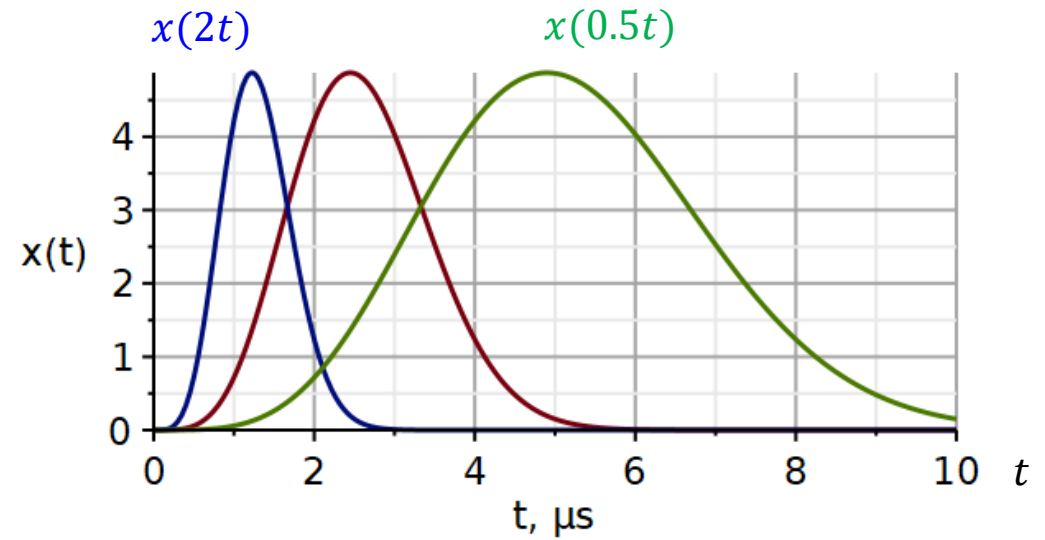
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Aperiodic signal:
$$x(t) = \begin{cases} 0 & t < 0 \\ t^4 e^{-t^2/3} & t \geq 0 \end{cases}$$

Time scaling:
$$y(t) = x(a \cdot t)$$

Time shift:
$$y(t) = x(t - t_0)$$



1.3.4 Combined operations. Order of operations is important

Scaling → *Shift*

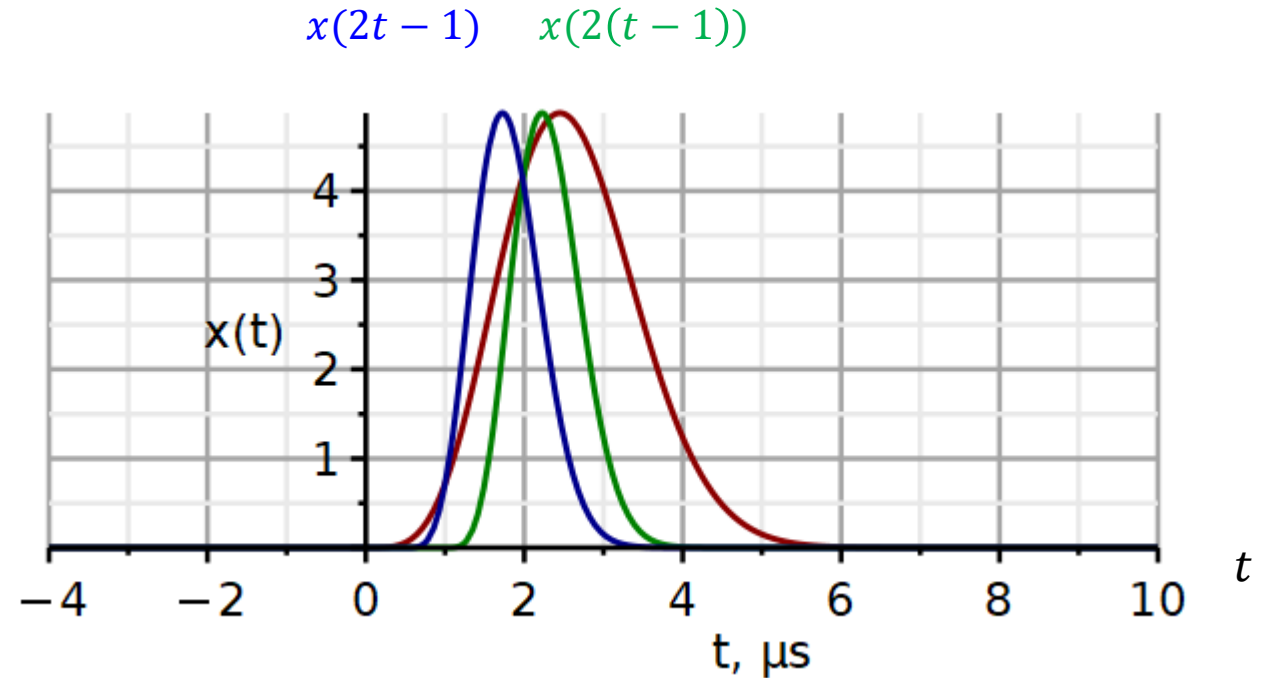
$$s(t) = x(at)$$

$$y(t) = s(t - t_0) = x(a(t - t_0))$$

Shift → *Scaling*

$$s(t) = x(t - t_0)$$

$$y(t) = s(at) = x(at - t_0)$$

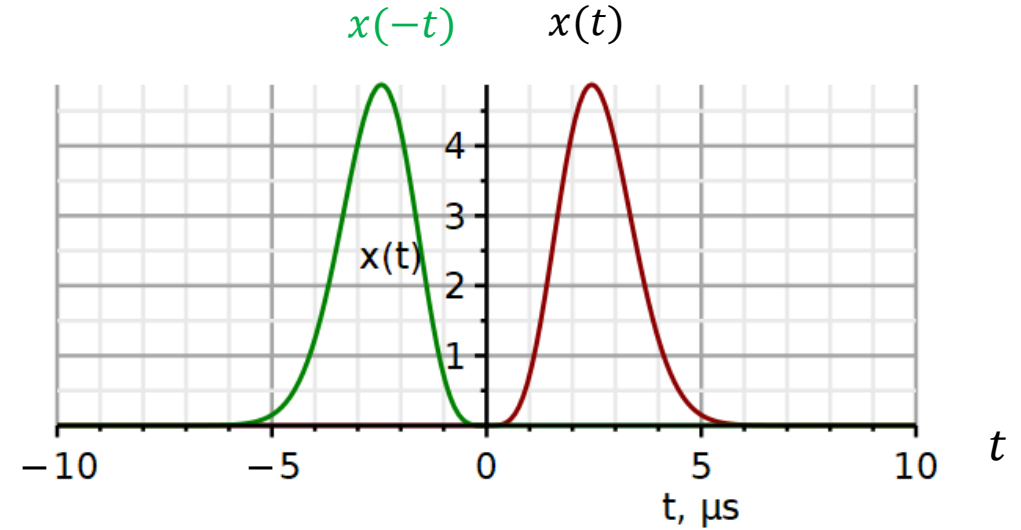


Aperiodic signal:

$$x(t) = \begin{cases} 0 & t < 0 \\ t^4 e^{-t^2/3} & t \geq 0 \end{cases}$$

Time reflection:

$$y(t) = x(-t)$$



Shift → Reflect

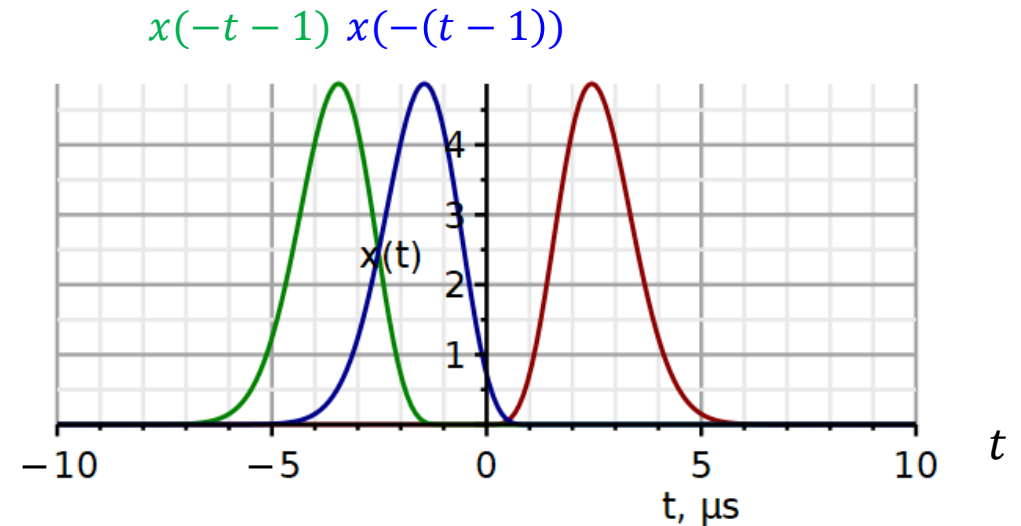
$$s(t) = x(t - t_0)$$

$$y(t) = s(-t) = x(-t - t_0)$$

Reflect → Shift

$$s(t) = x(-t)$$

$$y(t) = s(t - t_0) = x(-(t - t_0))$$

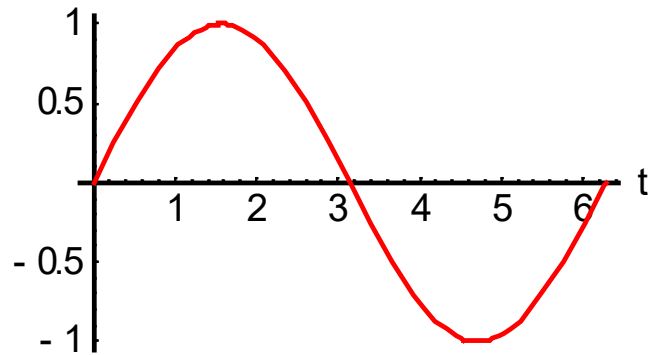


Signal energy:

$$E_x \stackrel{\text{def}}{=} \int_{-\infty}^{\infty} |x(t)|^2 dt$$

Example:

One Sine Period



$$x(t) = \begin{cases} 0 & , \quad t < 0 \\ A \sin(\omega t) & , \quad 0 \leq t \leq 2\pi/\omega \\ 0 & , \quad t > 2\pi/\omega \end{cases}$$

$$\begin{aligned} E_x &= \int_0^{2\pi/\omega} A^2 \sin^2(\omega t) dt = \frac{A^2}{2} \int_0^{2\pi/\omega} [\cos(\omega t - \omega t) - \cos(\omega t + \omega t)] dt \\ &= \frac{A^2}{2} \int_0^{2\pi/\omega} [1 - \cos(2\omega t)] dt = \frac{A^2}{2} \frac{2\pi}{\omega} = \frac{A^2}{2} T \end{aligned}$$

Signal power:

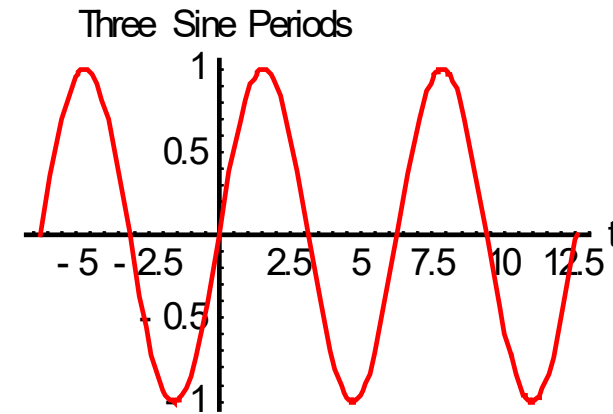
For a periodic signal, the energy is finite in every period. Thus, the total energy in all periods is infinite. Instead, we can define its power as the energy in one period divided by the period duration.

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

$$P_x = \frac{1}{T_0} \int_{t_1}^{t_1+T_0} |x(t)|^2 dt$$

$$P_x \stackrel{\text{def}}{=} \lim_{T_0 \rightarrow \infty} \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} |x(t)|^2 dt$$

$$x(t) = \sin(t)$$



$$P_x = \frac{E_{x,1p}}{T_0} = \frac{\frac{A^2}{2} \frac{2\pi}{\omega}}{\frac{2\pi}{\omega}} = \frac{A^2}{2}$$

$$x_{rms} = \sqrt{P_x} = \frac{A}{\sqrt{2}} \approx 0.7071 \cdot A$$

- Energy signal: A signal with finite energy
- Power signal: A signal with finite power
- A periodic signal with finite energy in one period will have infinite energy in all periods
- Thus a periodic signal is never an energy signal.
- Signals with infinite power is neither an energy nor a power signal

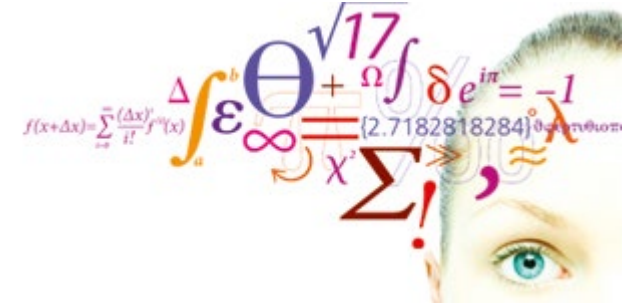
Measure of deviation:

Mean Square Error

$$MSE \stackrel{\text{def}}{=} \int_{-\infty}^{\infty} |x(t) - \tilde{x}(t)|^2 dt$$

Energy of error signal

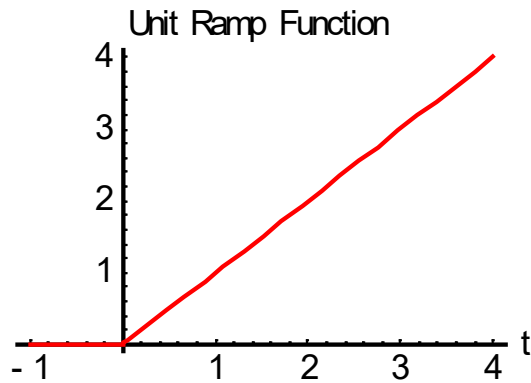
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Special signals

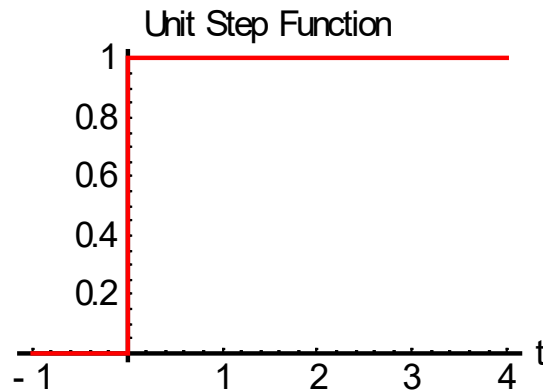
Unit ramp:

$$r(t) \stackrel{\text{def}}{=} \begin{cases} 0 & ,t < 0 \\ t & ,t \geq 0 \end{cases}$$



Unit step:

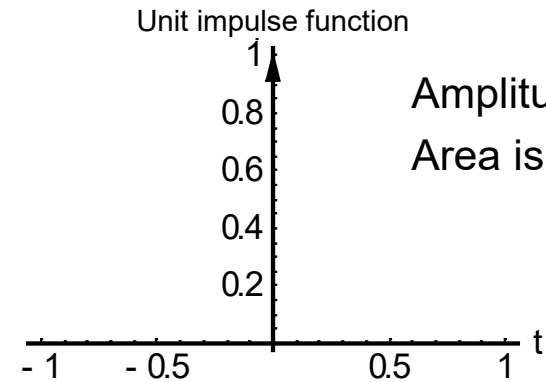
$$u(t) \stackrel{\text{def}}{=} \begin{cases} 0 & ,t < 0 \\ 1 & ,t > 0 \end{cases}$$



Unit impulse:

$$\delta(t) = 0, t \neq 0$$

$$\int_{-\infty}^{\infty} \delta(t) dt = 1$$



Amplitude is infinite, not 1.
Area is 1.

$$u(t) = \frac{dr}{dt}, r(t) = \int_{-\infty}^t u(\tau) d\tau$$

$$\delta(t) = \frac{du}{dt}, u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Properties of Impulse Function

Property 1: $\delta(t) = \delta(-t)$

Even function

Property 2: $\int_{0_-}^{0_+} A\delta(t) dt = A$

The area under the impulse function equals its strength

Property 3: $A\delta(t - t_0) = 0, t \neq t_0$

The impulse function is zero except at $t = t_0$

Property 4: $A\delta(t - t_0) + B\delta(t - t_0) = (A + B)\delta(t - t_0)$

Superimposed impulse functions add their strengths

Property 5: $y(t)\delta(t - t_0) = y(t_0)\delta(t - t_0)$

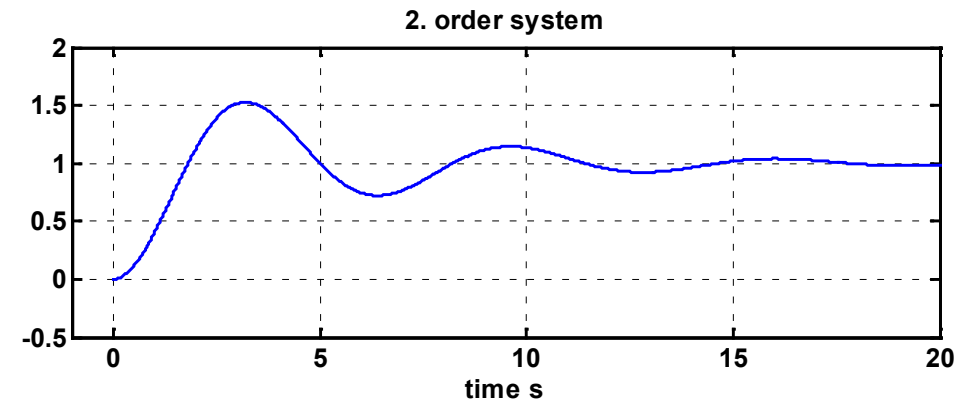
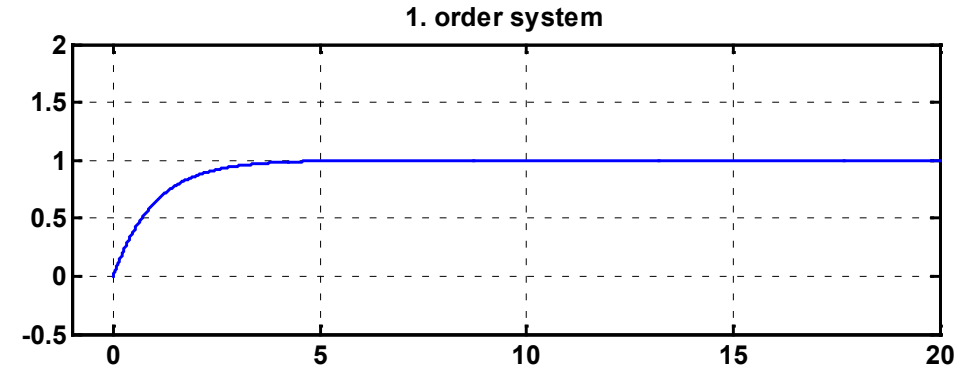
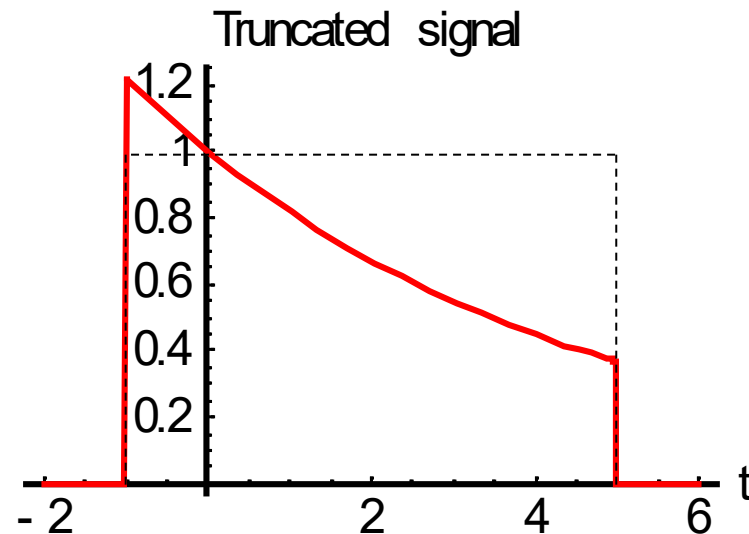
Multiplication of a function by an impulse function yields an impulse function with strength equal the function amplitude at the impulse location

Application of step function

1: As a test signal to analyze the "Step Response" of a system



2: Truncation of signals



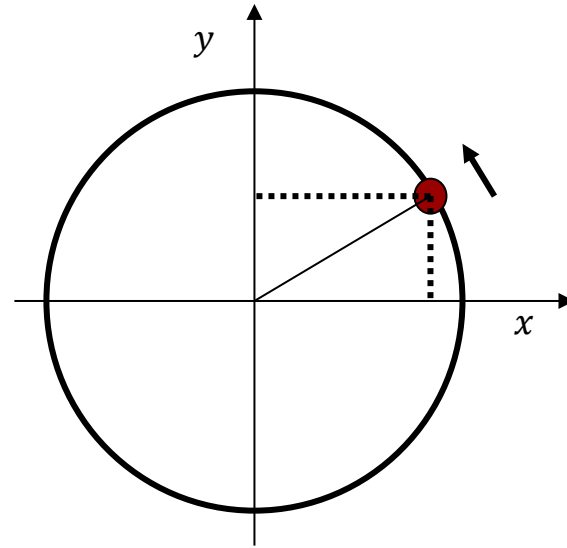
$$x(t) = e^{-t/5} [u(t+1) - u(t-5)]$$

Sinusoidal signals

$$x(t) = A \cdot \cos[\theta(t)]$$

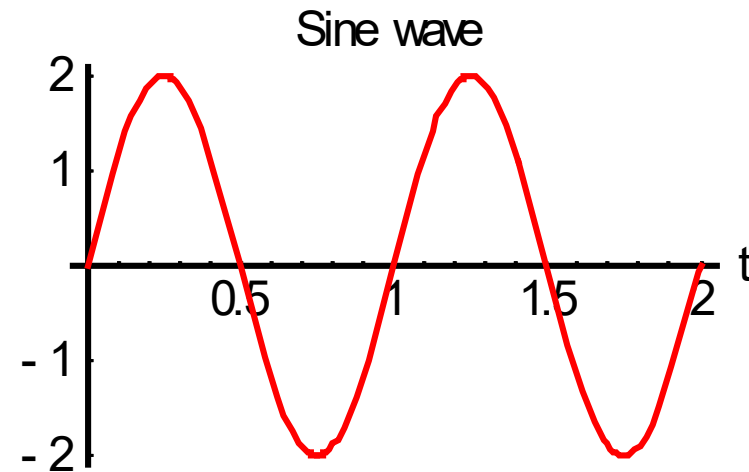
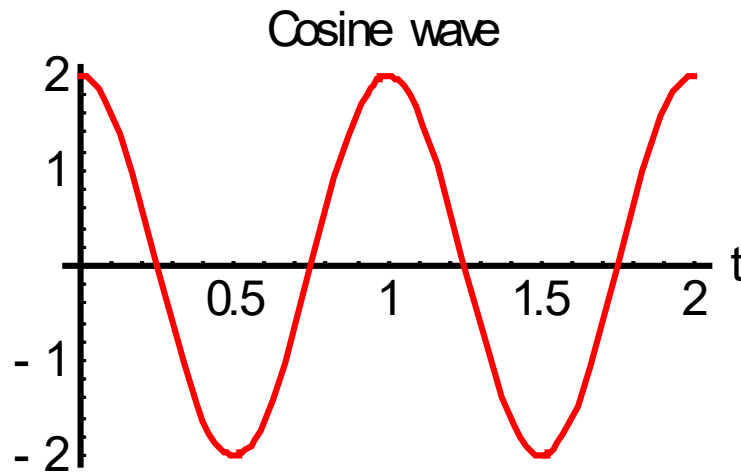
$$x(t) = A \cdot \cos[2\pi \cdot f \cdot t + \theta_0]$$

$$y(t) = A \cdot \sin[2\pi \cdot f \cdot t + \theta_0]$$

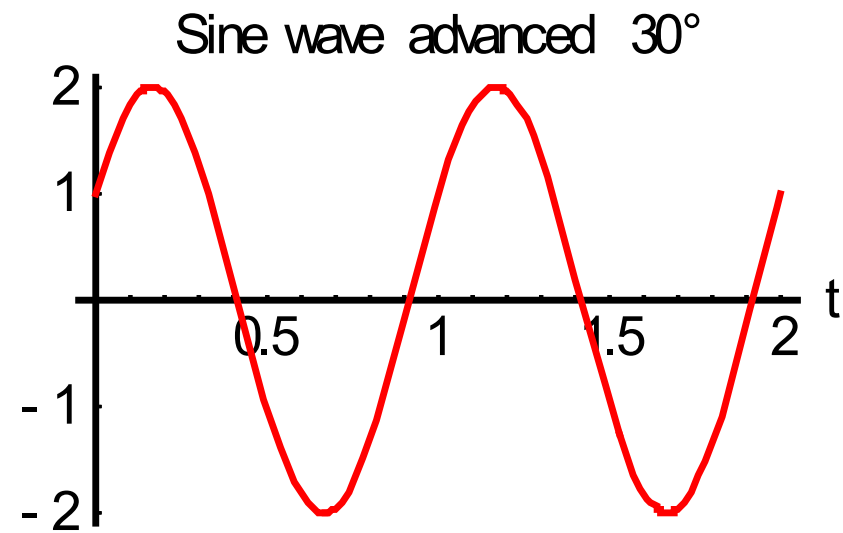
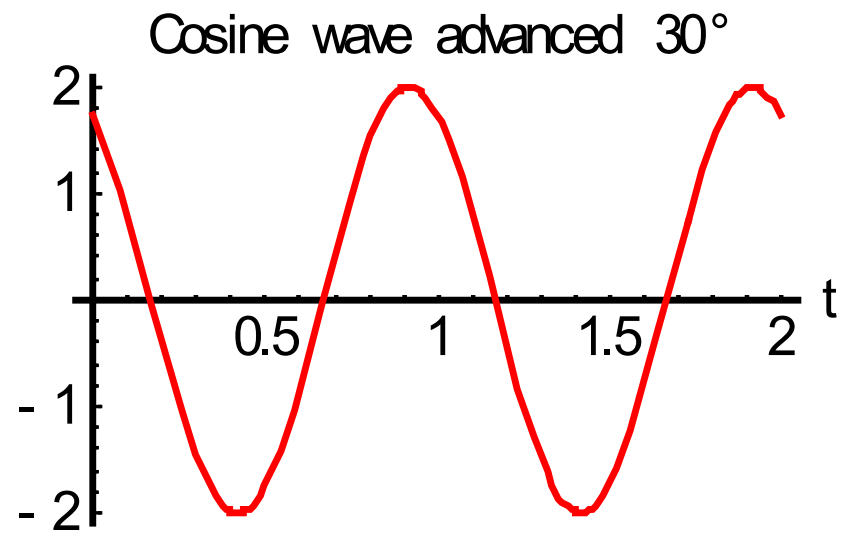
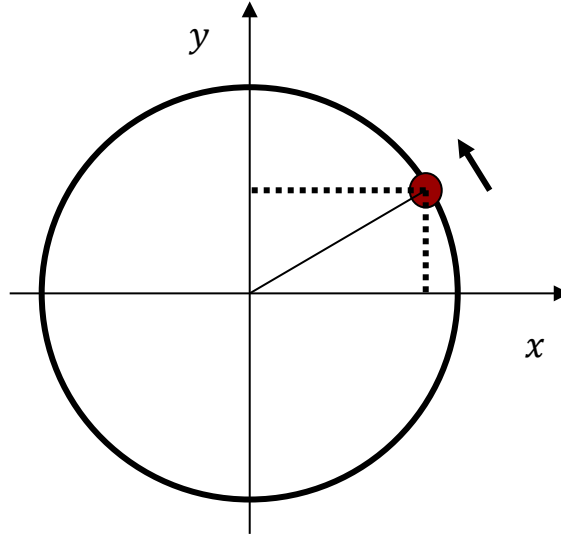


$$\theta(t) = \frac{d\theta}{dt} \cdot t = \omega \cdot t$$

$$\omega = \frac{2\pi}{T_0} = 2\pi \cdot f$$

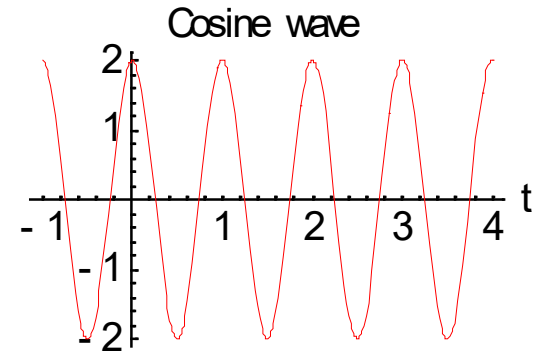


Phase angle

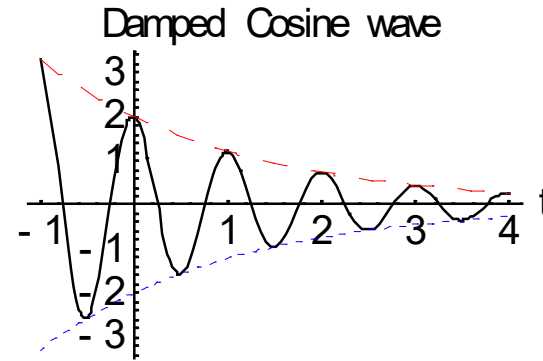


Exponentially varying sinusoidals

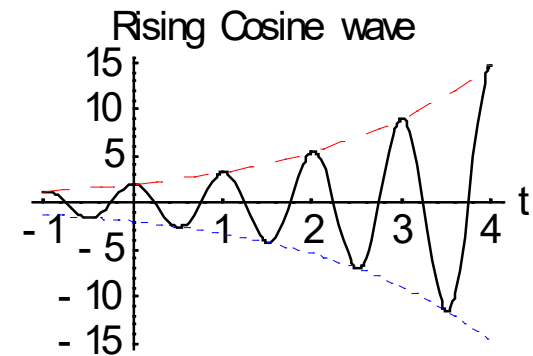
$$x_1(t) = A \cdot \cos[2\pi \cdot f \cdot t]$$



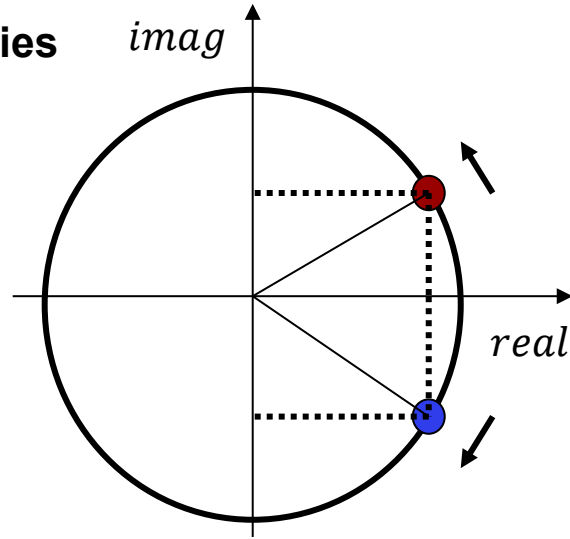
$$x_2(t) = \underbrace{A \cdot e^{-t/\tau}}_{\text{envelope}} \cdot \cos[2\pi \cdot f \cdot t]$$



$$x_3(t) = \underbrace{A \cdot e^{t/\tau}}_{\text{envelope}} \cdot \cos[2\pi \cdot f \cdot t]$$



Euler's identities



$$e^{j\omega t} = \cos(\omega t) + j \sin(\omega t)$$
$$e^{-j\omega t} = \cos(\omega t) - j \sin(\omega t)$$

$$e^{j\omega t} + e^{-j\omega t} = 2 \cos(\omega t)$$
$$e^{j\omega t} - e^{-j\omega t} = j2 \sin(\omega t)$$

As a mathematical abstraction, we can represent any sinusoidal oscillation as the sum of two rotations, one with a positive frequency, and one with an identical negative frequency.

From a physical perspective, oscillations have only positive frequencies.

Why would we ever use complex exponentials rather than cos and sin?

If we insert a cos function into a differential equation, we obtain a sum of cos and sine functions, which are hard to add up. If we insert a complex exponential function, we obtain a sum of only complex exponential functions.

Complex exponential:

$$\begin{aligned} z(t) &= e^{\sigma t} [\cos(\omega t) + j \sin(\omega t)] \\ &= e^{\sigma t} e^{j\omega t} = e^{(\sigma + j\omega)t} \\ &= e^{st}, s \stackrel{\text{def}}{=} \sigma + j\omega \end{aligned}$$

A complex exponential has one parameter s .
 $\text{Re}(s)$ describes the exponential change in amplitude.
 $\text{Im}(s)$ describes the frequency of oscillation.

Complex conjugated:

$$\begin{aligned} z^*(t) &= e^{\sigma t} [\cos(\omega t) - j \sin(\omega t)] \\ &= e^{\sigma t} e^{-j\omega t} = e^{(\sigma - j\omega)t} \\ &= e^{s^*t}, s^* \stackrel{\text{def}}{=} \sigma - j\omega \end{aligned}$$

$$z(t) = x(t) + jy(t) \quad z^*(t) = x(t) - jy(t)$$

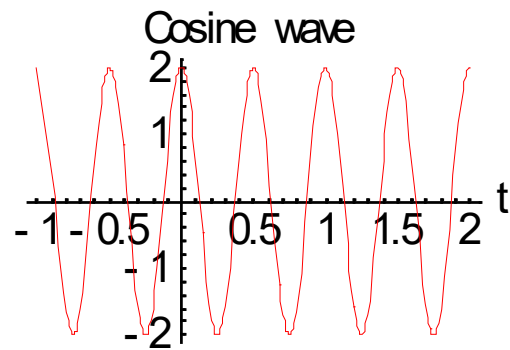
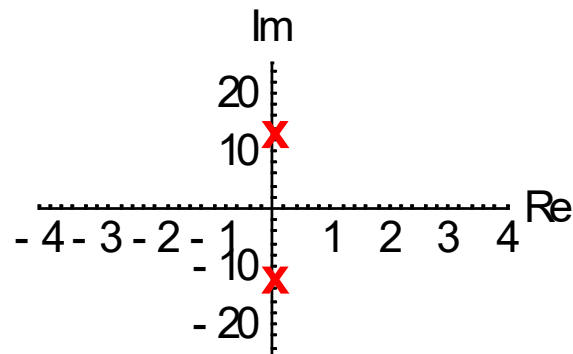
Even function:

$$x(t) = \frac{1}{2} (z(t) + z^*(t)) = \frac{1}{2} (e^{st} + e^{s^*t}) = e^{\sigma t} \cos(\omega t)$$

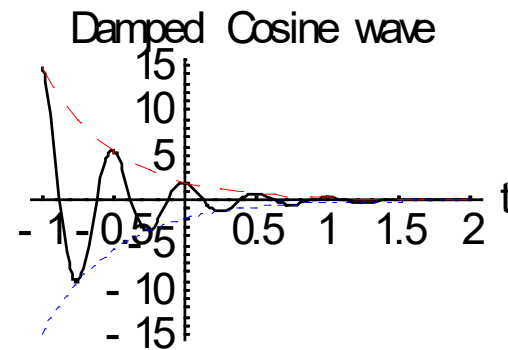
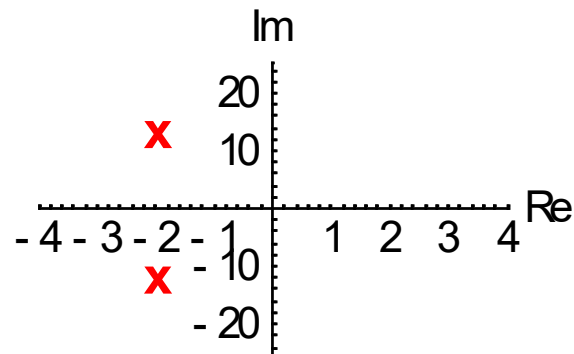
Odd function

$$y(t) = \frac{1}{2} (z(t) - z^*(t)) = \frac{1}{2} (e^{st} - e^{s^*t}) = e^{\sigma t} \sin(\omega t)$$

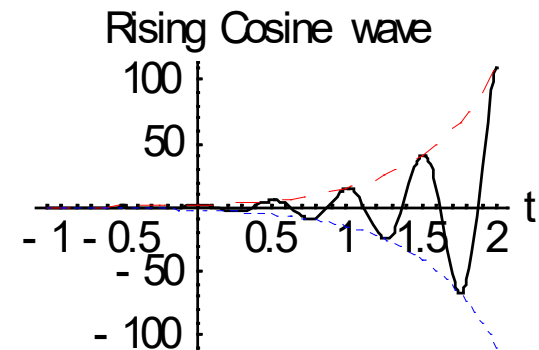
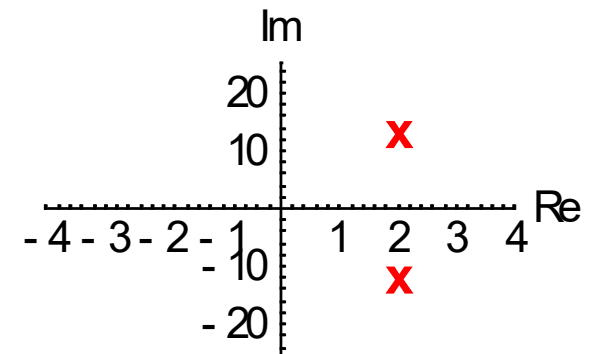
$$s = j2\pi \cdot 2$$



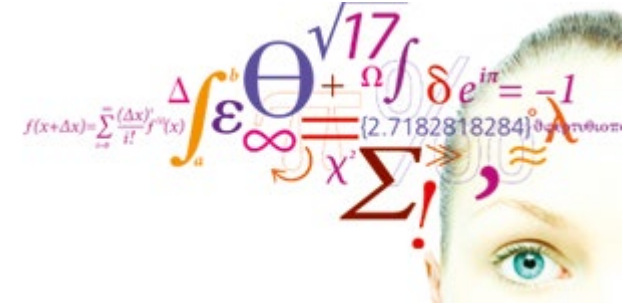
$$s = -2 + j2\pi \cdot 2$$



$$s = 2 + j2\pi \cdot 2$$



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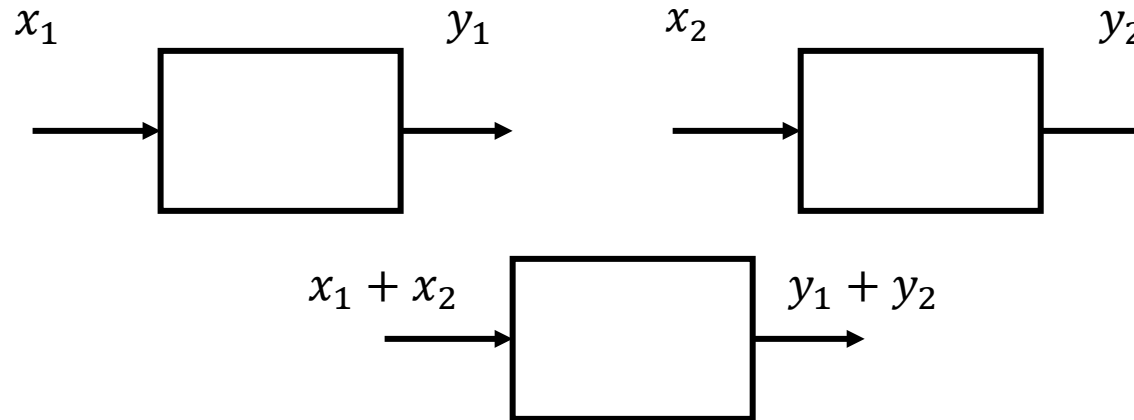


Video 3

Linear systems

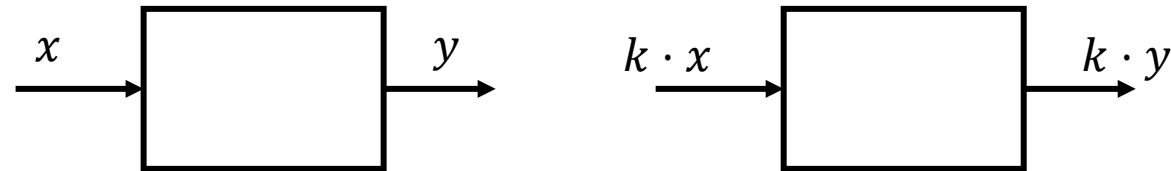
Additive property:

+



Homogeneity property:

=



Superposition property:



A system is linear if and only if the superposition property holds.



$$x(t) = k_1 x_1(t) + k_2 x_2(t)$$

$$k_1 \ddot{y}_1(t) + a_1 k_1 \dot{y}_1(t) + a_0 k_1 y_1(t) = b_0 k_1 x_1(t)$$

$$y(t) = k_1 y_1(t) + k_2 y_2(t)$$

$$k_2 \ddot{y}_2(t) + a_1 k_2 \dot{y}_2(t) + a_0 k_2 y_2(t) = b_0 k_2 x_2(t)$$

$$\dot{y}(t) = k_1 \dot{y}_1(t) + k_2 \dot{y}_2(t)$$

$$\ddot{y}(t) = k_1 \ddot{y}_1(t) + k_2 \ddot{y}_2(t)$$

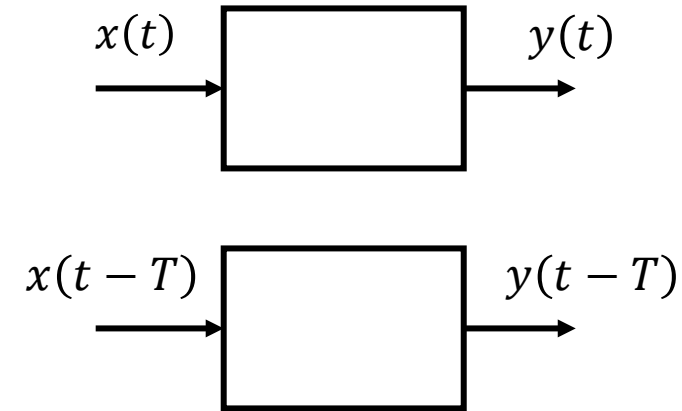
$$\underbrace{k_1 \ddot{y}_1(t) + k_2 \ddot{y}_2(t)}_{\ddot{y}(t)} + a_1 \underbrace{(k_1 \dot{y}_1(t) + k_2 \dot{y}_2(t))}_{\dot{y}(t)} + a_0 \underbrace{(k_1 y_1(t) + k_2 y_2(t))}_{y(t)} = b_0 \underbrace{(k_1 x_1(t) + k_2 x_2(t))}_{x(t)}$$

$$\ddot{y}(t) + a_1 \dot{y}(t) + a_0 y(t) = b_0 x(t)$$

If a system has a linear differential equation, it is a linear system.

Time-Invariant systems

A system is time-invariant if a time-shifted input yields a correspondingly time-shifted output.



A system is time-invariant if the system itself doesn't vary with time. This holds if all system parameters are constants.

The differential equation for a time-invariant system must have **constant coefficients**.

Time-invariance **refers ONLY to the system, NOT the signals**. For a time-invariant system, the input and output signals can vary as functions of time.

LTI System = Linear Time-Invariant System

Causal Systems

A system is causal if its output depends only on present and past input.

A noncausal system can have output that depend on future input. It must predict future values of the input signal. A noncausal system can have an output before the input is applied, i.e., without "cause".

Physical systems are always causal.

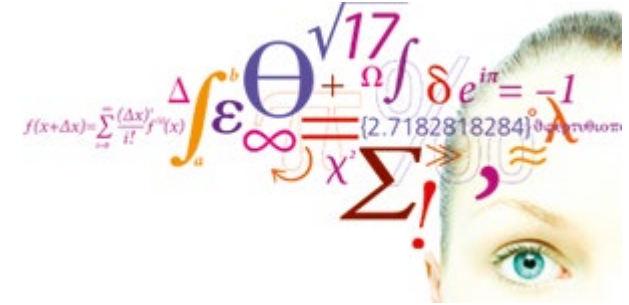
Noncausal systems are only of theoretical interest.

Which system is causal?

$$A: y(t) = \int_{t-5}^{t+5} x(\tau) d\tau$$
$$B: y(t) = \int_{t-5}^t x(\tau) d\tau$$

LTIC System = Linear Time-Invariant Causal System

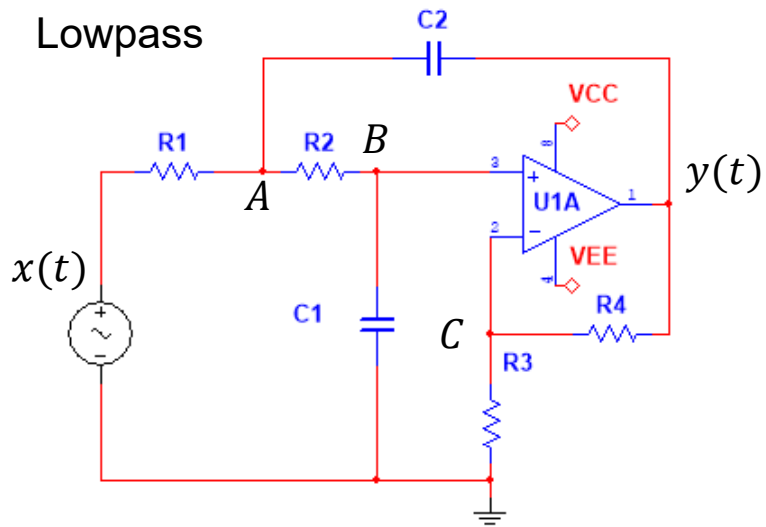
- Lecture
 - Overview of "Signals and Systems part"
 - Classification of signals
 - Operations on signals
 - Special signals
 - Classification of systems
 - **Overview of filter problems**
- Exercises



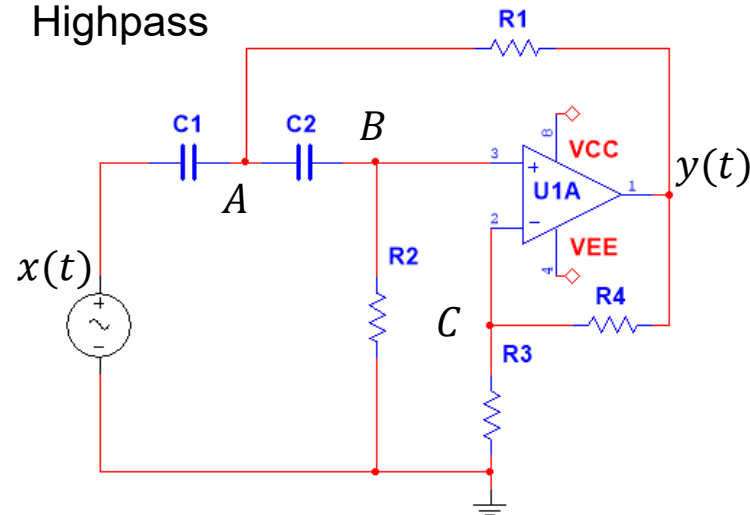
Being able to analyze the performance of filter circuits and to design new filter circuits meeting given design specifications is a very valuable competency for students in both Electrotechnology and in Biomedical engineering.

This course will use Sallen-Key filter circuits as a common platform for application of theories and methods.

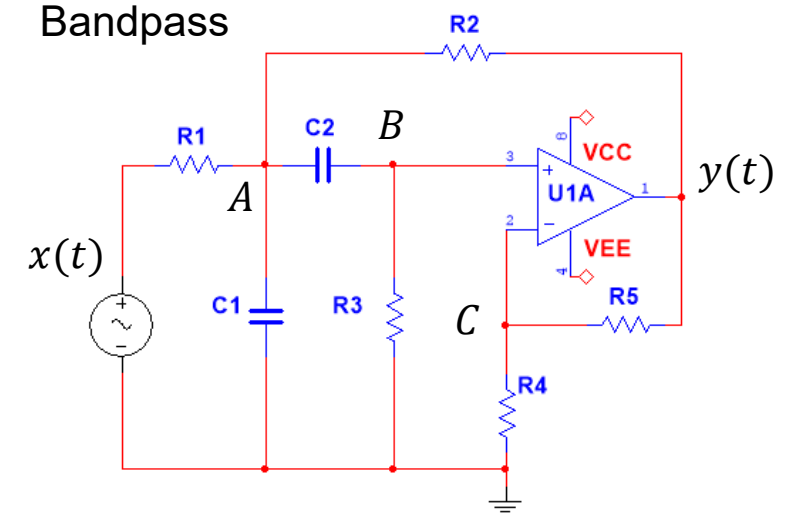
Lowpass



Highpass



Bandpass



All three circuits can be modelled by this differential equation: $\ddot{y} + a_1\dot{y} + a_0y = b_2\ddot{x} + b_1\dot{x} + b_0x$

Why is this a lowpass filter?

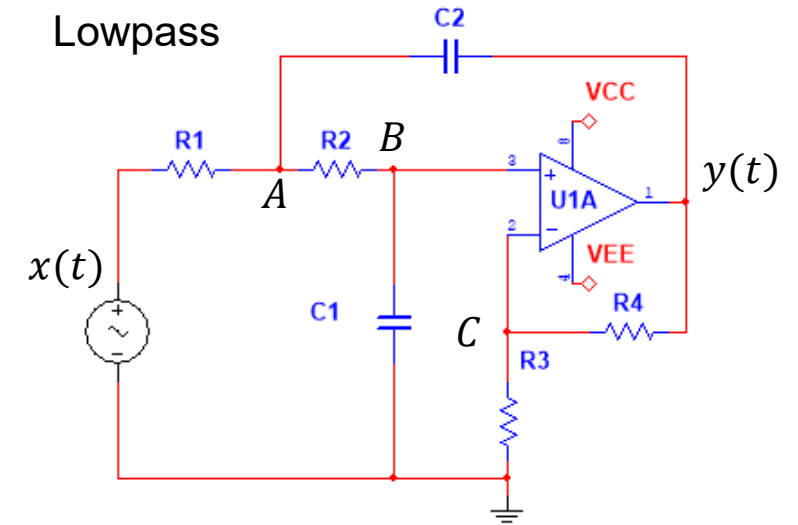
At high frequencies, C_1 will become a short circuit. The voltage V_B will approach 0V and $y(t) = KV_B(t)$ will also approach zero.

Problem 1

$$K = 1 + \frac{R_4}{R_3}$$

$$a_1 = \frac{1}{R_1 C_2} + \frac{1}{R_2 C_2} + \frac{1 - K}{R_2 C_1} \quad a_0 = \frac{1}{R_1 R_2 C_1 C_2}$$

$$b_2 = 0 \quad b_1 = 0 \quad b_0 = \frac{K}{R_1 R_2 C_1 C_2}$$



$$\ddot{y} + a_1 \dot{y} + a_0 y = b_0 x$$

The differential equation for a highpass filter

Why is this a highpass filter?

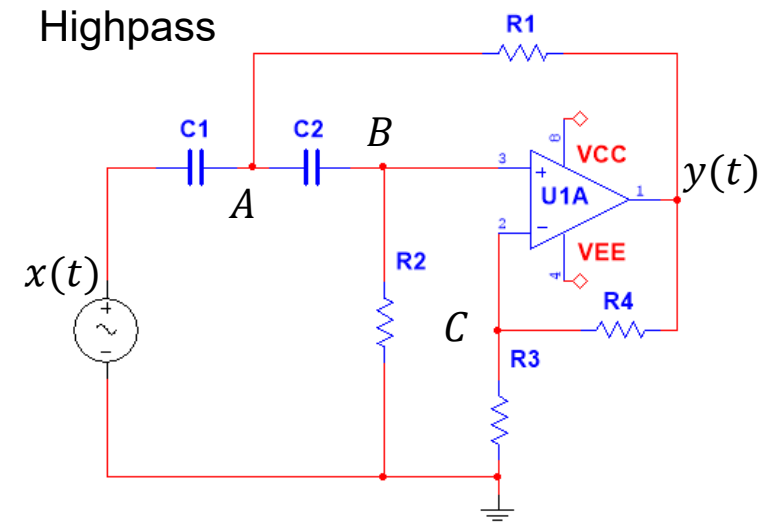
At low frequencies, C_1 and C_2 will block currents. As the current through R_2 approaches zero, the voltage V_B will approach 0V and $y(t) = KV_B(t)$ will also approach 0V.

Problem 2

$$K = 1 + \frac{R_4}{R_3}$$

$$a_1 = \frac{1}{R_2 C_2} + \frac{1}{R_2 C_1} + \frac{1 - K}{R_1 C_1} \quad a_0 = \frac{1}{R_1 R_2 C_1 C_2}$$

$$b_2 = K \quad b_1 = 0 \quad b_0 = 0$$



$$\ddot{y} + a_1 \dot{y} + a_0 y = b_2 \ddot{x}$$

The differential equation for a bandpass filter

Why is this a bandpass filter?

At low frequencies, C_2 will block currents. The voltage V_B will approach 0V and $y(t) = KV_B(t)$ will also approach zero.

At high frequencies, C_1 will short circuit. The voltage V_B will approach 0V and $y(t) = KV_B(t)$ will also approach zero.

Problem 3

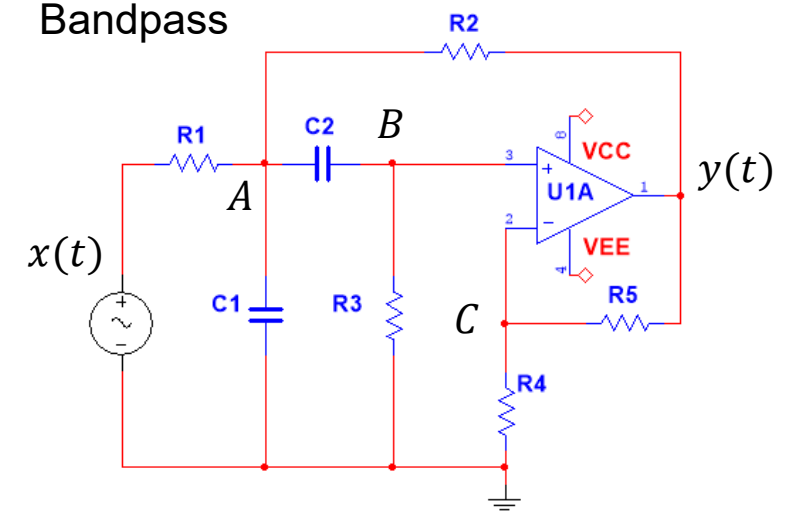
$$K = 1 + \frac{R_4}{R_3}$$

$$a_1 = \frac{1}{R_1 C_1} + \frac{1}{R_3 C_2} + \frac{1}{R_3 C_1} + \frac{1 - K}{R_2 C_1}$$

$$a_0 = \left(\frac{1}{R_1} + \frac{1}{R_2} \right) \frac{1}{R_3 C_1 C_2}$$

$$b_2 = 0 \quad b_1 = K \frac{1}{R_1 C_1} \quad b_0 = 0$$

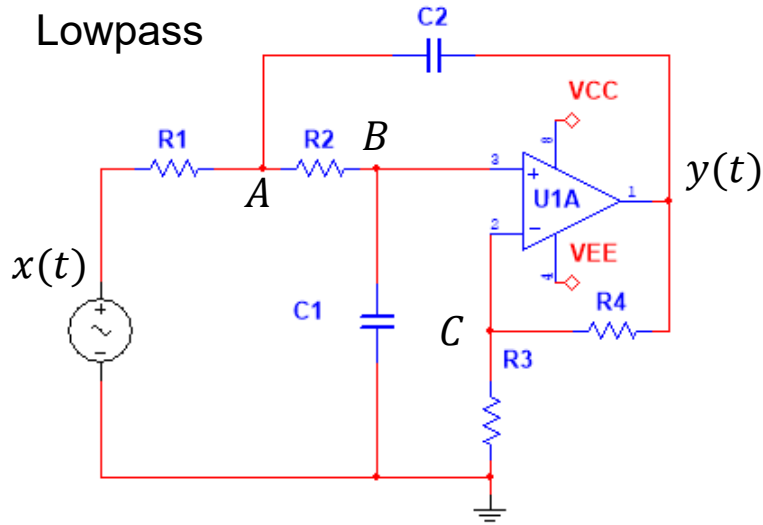
Bandpass



$$\ddot{y} + a_1 \dot{y} + a_0 y = b_1 \dot{x}$$

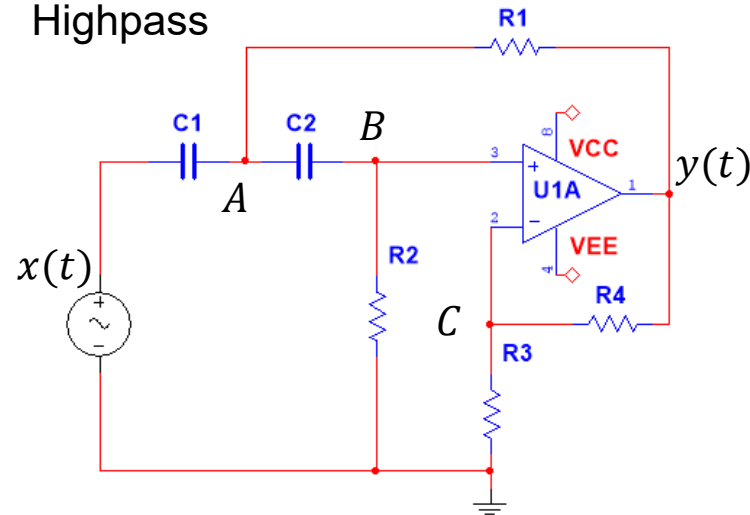
Filter circuits used in this course.

Lowpass



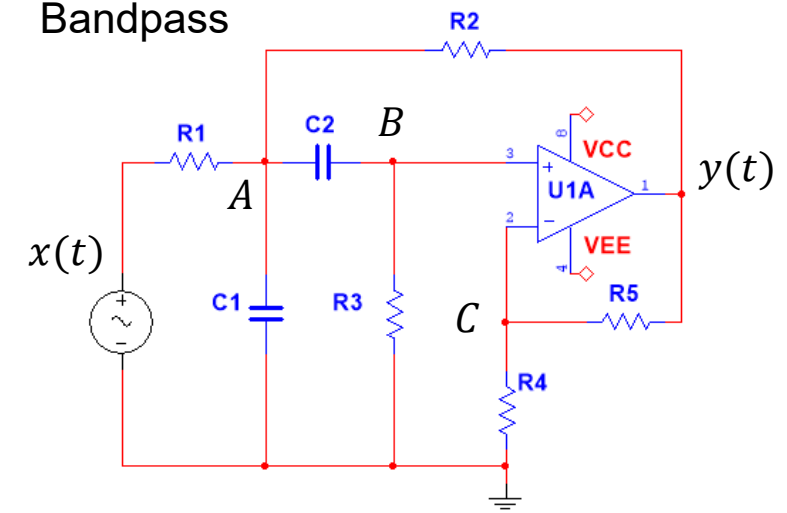
$$\ddot{y} + a_1\dot{y} + a_0y = b_0x$$

Highpass



$$\ddot{y} + a_1\dot{y} + a_0y = b_2\ddot{x}$$

Bandpass

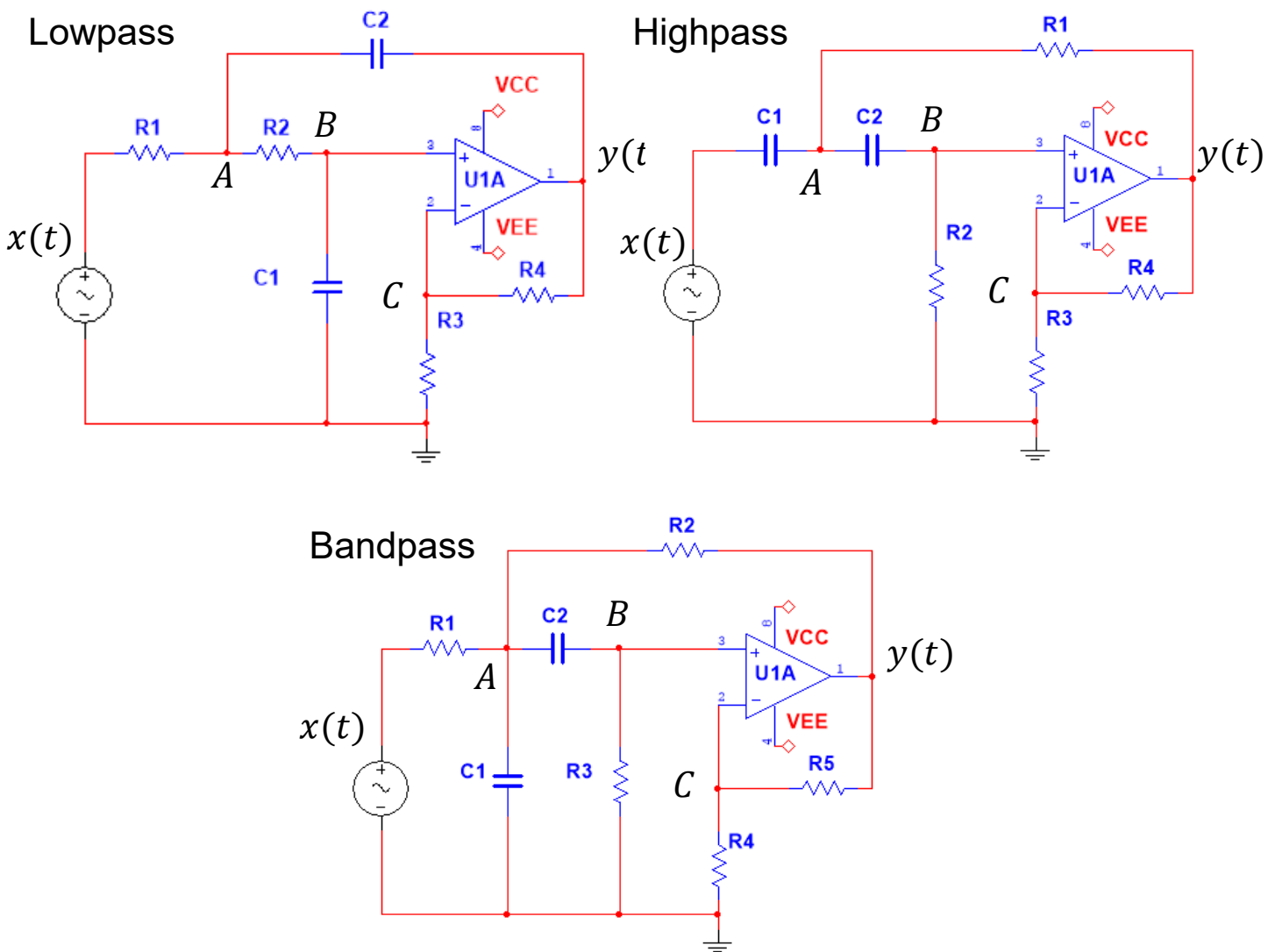


$$\ddot{y} + a_1\dot{y} + a_0y = b_1\dot{x}$$

Observation:

- Same left-hand side. Different right-hand side, hence different particular solution.
- The characteristic equation can have one of three different solutions: two real roots, one double root, a complex conjugated pair.
- The three filters above have component values that will cover all of the three possibilities. Although the filter problems look identical, the techniques used to solve them differ because of the difference in the characteristic equations.

Sallen-Key filters used in this course



Filter table

		Lowpass	Highpass	Bandpass
R_1	$k\Omega$	3.9894	1784.1	56.419
R_2	$k\Omega$	0.8865	892.06	34.117
R_3	$k\Omega$	1.0	1.0	149.72
R_4	$k\Omega$	1.0	1.0	1.0
R_5	$k\Omega$	—	—	1.0
C_1	nF	1795.2	1784.1	56.419
C_2	nF	398.94	3568.2	56.419
a_1		$2.83 \cdot 10^3$	$6.283 \cdot 10^{-1}$	$3.141 \cdot 10^1$
a_0		$3.95 \cdot 10^5$	$9.87 \cdot 10^{-2}$	$9.87 \cdot 10^4$
b_2		0	2	0
b_1		0	0	$6.283 \cdot 10^2$
b_0		$7.89 \cdot 10^5$	0	0